

RAVIBALAN CHITHAMPARAVEL

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PROFESSIONAL SUMMARY:

Data Engineer and Database Developer with 3.5+ years of experience in product-based company environments, specializing in designing, developing, and optimizing enterprise-grade data and backend systems using Microsoft SQL Server, PostgreSQL, MongoDB, Node.js, and Express.js. Strong expertise in T-SQL development, including stored procedures, query optimization, indexing, performance tuning, and ETL-based data integration for high-volume transactional systems. Experienced in building scalable data pipelines, ensuring data accuracy, system reliability, and production support efficiency. Proven leadership experience as a Team Lead, mentoring developers, enforcing coding standards, and collaborating with cross-functional teams to deliver high-performance, data-driven solutions.

TECHNICAL SKILLS :

- **Databases:** Microsoft SQL Server, PostgreSQL, MongoDB, Elasticsearch
- **Data Engineering & ETL:** ETL Development, Data Integration, Data Pipeline Development, Data Modeling, Data Reconciliation, Batch Processing, Apache Airflow, SSIS, Pandas, Apache Spark, PySpark
- **SQL Development:** T-SQL, Stored Procedures, User-Defined Functions (UDFs), Triggers, Views, CTEs, Joins, Window Functions, Temporary Tables, Dynamic SQL
- **Performance Optimization:** Query Optimization, Execution Plan Analysis, Indexing (Clustered & Non-Clustered), Database Performance Tuning, Query Debugging, High-Volume Data Processing
- **Backend & APIs:** Node.js, Express.js, REST APIs, Axios, Microservices, Prisma ORM
- **Cloud & Data Platforms:** Microsoft Azure, Azure AI Custom Document Extraction, Azure Blob Storage, Azure Functions, Serverless APIs
- **Programming Languages:** Python, JavaScript, T-SQL
- **Messaging & Distributed Systems:** RabbitMQ
- **DevOps & Tools:** Git, Docker, SSMS, SSRS, PgAdmin

EXPERIENCE & KEY PROJECTS:

GOVE TECHNOLOGIES | NOV 2022 - PRESENT

DATA ENGINEER [BACKEND & AI AUTOMATION] | NOV 2025 - MAR 2026

AI-Powered Document Intelligence & OCR Automation Platform

- Designed and implemented a Multi-Tenant AI-powered document processing platform using Microsoft Azure, Azure AI Custom Document Extraction, Azure Blob Storage, Python, Pandas, OCR, and Serverless APIs for automated extraction and validation of invoices, toll receipts, permits, and financial documents.
- Trained and optimized Azure AI Custom Document Extraction models using 100+ document and image formats with varying layouts, structures, and vendor-specific templates.
- Built scalable serverless APIs for document ingestion, AI processing orchestration, OCR trigger workflows, and processed result management.
- Implemented secure document upload and storage workflows using Azure Blob Storage for handling high-volume document and image processing operations.
- Automated OCR-based data extraction, field mapping, transformation, and validation for complex semi-structured and unstructured document layouts.
- Developed post-processing pipelines using Python and Pandas for data cleansing, normalization, validation checks, duplicate detection, and exception analysis before database ingestion.
- Designed automated error-handling and validation mechanisms to improve extraction reliability and reduce manual intervention during document processing workflows.
- Integrated extracted document data into backend billing and reconciliation systems for downstream transactional processing and audit workflows.
- Collaborated with backend and business teams to improve document classification accuracy, extraction consistency, and operational scalability.
- Reduced manual document processing effort by 40% while improving extraction accuracy and processing efficiency across multiple document types.

DATA ENGINEER [BACKEND] | DEC 2022 – OCT 2025

Multi Tenant Fleet And Toll Management Platform

- Designed and developed scalable architecture and data processing systems for multi-tenant enterprise billing, payment, and transaction processing platforms using Node.js, Microsoft SQL Server, and PostgreSQL.
- Led a 6-member backend team, managing task planning, code reviews, technical guidance, deployment support, and production issue resolution.
- Designed IAM-based data models and access management systems to securely manage multi-tenant vendor, billing, transactional, and document-level access controls.
- Built secure RESTful APIs and microservices for billing workflows, transaction processing, reconciliation, and audit operations across multi-tenant environments.
- Developed and optimized complex T-SQL components including stored procedures, functions, triggers, indexing strategies, query optimization, and reverse engineering of existing database structures for high-volume transactional systems handling millions of records.
- Managed and processed 50GB+ of fleet, toll, payment, and transactional datasets using Microsoft SQL Server.
- Implemented ETL-based data pipelines, workflow automation, data validation frameworks, reconciliation processes, and exception-handling mechanisms at both database and application layers.
- Designed and implemented Medallion Architecture-based data processing workflows for structured transactional data transformation and validation.
- Implemented Slowly Changing Dimension (SCD) concepts and late-arriving data handling strategies for maintaining historical and incremental transactional data consistency.
- Designed automated reconciliation pipelines for vendor billing, bank transactions, and toll processing systems, significantly reducing manual verification efforts and operational errors.
- Used Python with Pandas for transactional data analysis, data cleansing, normalization, exception reporting, and trend analysis.
- Developed and maintained Apache Airflow workflows for transaction ingestion, billing automation, scheduled ETL jobs, and data pipeline orchestration.
- Implemented Amazon S3-based data lake storage for multi-tenant toll-billed images, vendor documents, acknowledgements, and transactional file management, enabling scalable and secure document storage and retrieval workflows.
- Implemented vendor-specific billing and reconciliation logic supporting 110+ U.S. toll vendors with complex business rules and financial processing requirements.
- Collaborated with cross-functional teams to deliver scalable, reliable, and high-performance backend and data solutions in production environments.
- Improved reconciliation turnaround time by 45% and reduced manual operational errors by 35% through workflow automation and performance optimization.

EDUCATION & CERTIFICATION

- Completed Advanced SQL Certifications in Hackerrank - [Advanced SQL Test Certification link](#)
- Bachelor Of Computer Application | Madras University - Expected Dec 2026
- Computer Engineering in Diploma | Mahendra Group Of Institutions - Jul 2022